



5/10th
5/12/2026

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Q.1) Explain how neurons are modelled in artificial neuron system. Explain role of weight, input, summation, act. function & output in artificial neuron.

Q.2) Explain the concepts of supervised & unsupervised learning in NN. How does the availability of target output influence the learning & adaptation process in each case.

Q.3)

Ans 1) 1. In an artificial neuron system, the neurons act as nodes.

2. Each neuron has 3 layers:

- Input layer
- Hidden layer(s)
- Output layer.

3. The input layer is responsible for the input data coming to the neuron. Input is crucial to a neuron as without it the neuron does not get activated.

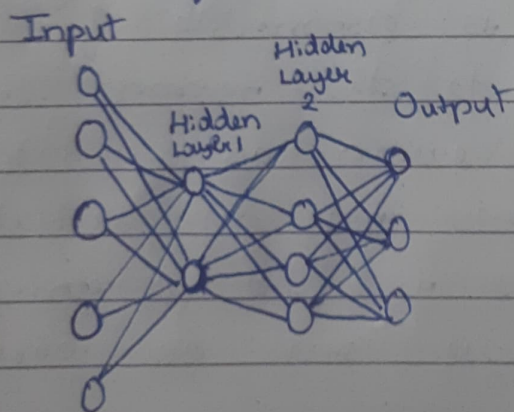
4. The Hidden layer(s) is the processing layer. There can be one or more hidden layers.

5. In the hidden layer following steps occur:

- Multiplication of weights & their summation ($\sum w_i x_i$)
- Addition of bias.



6. Weight is the strength that the bond between neurons have.
7. Each bond has a certain weight.
8. Bias is used to shift the curve of activation function. It is a predecided constant which is added in the hidden layer.
9. Activation function is used to introduce non-linearity in the function.
10. Without activation function the model will always remain linear. There are many different types of activation functions:
 - a) Sigmoid function.
 - b) Tanh function
 - c) ReLU function





Ans 2)

Supervised Learning

1. In this technique labelled data is given as the input.
2. In this method, the output is already known.
3. This method is great for tasks like classification.
4. E.g. Image classification, Spam detection.
5. The availability of target output helps predefine the classes. The model is taught what belongs to which class & this learning is then implemented on the new data.

Unsupervised Learning

1. In this technique, the input data is not labelled.
2. In this method, the output is not known.
3. This method is good for tasks like clustering.
4. E.g. Mineral hotspot clustering.
5. The target output is not known hence classes / clusters are not known. The model is taught to group the available data points based on similarity & then the new data point is assigned to a cluster / group.